

Seemingly Virtuous Complexity in Return Prediction

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Framework

Core Question: Does high-complexity RFF learn genuine return predictability, or mechanically reproduce a simple trading rule?

1: Ridgeless regression algebra. When $P > T$, the forecast must satisfy $\hat{r}_{t+1|t}^{rff} = w_t' r_t$: a weighted average of the past T training-window returns.

2: RFF–kernel equivalence. RFF dot products approximate Gaussian kernels, so w_t reflects similarity between current predictors x_t and past predictors x_{t-k} .

3: Empirical reconstruction. Rebuilding the RFF strategy and compares it with a kernel strategy and a simple volatility-timed momentum strategy.

4: Mechanism tests. Spanning tests, artificial return reversals, and wild-bootstrap predictors test whether RFF returns are explained by the mechanical strategy.

Implication: Strong OOS performance reflects mechanically generated volatility-timed momentum, not clear evidence of “virtuous complexity.”

Overparameterized Models

- A model is **overparameterized** when its number of effective predictors or parameters exceeds the number of training observations:

$$P \gg T.$$

- In classical statistics, this is usually viewed as dangerous because the model can perfectly fit training noise.
- Modern machine learning shows a more subtle possibility:

perfect in-sample fit $\nRightarrow$ poor out-of-sample performance.

- Kelly, Malamud, and Zhou (KMZ, 2024) report a striking result:

$$K = 15, \quad P = 12,000, \quad T = 12.$$

- This paper asks whether the strong performance comes from **genuine learning of return predictability** or from a **mechanical effect of overparameterized RFF regression**.

Motivation I: Why Is the KMZ Result a Puzzle?

1. The idea 'overparameterized' has entered empirical asset pricing through Random Fourier Features (RFF), which transform a small set of predictors into tons of nonlinear features.
2. In KMZ's setting:
 - Their RFF model generates strong out-of-sample market-timing performance.
 - This is surprising because aggregate stock returns are very noisy and many Goyal–Welch predictors are highly persistent.
3. If the result reflects genuine learning, it would challenge the conventional view that market return predictability requires long samples.
 - There may be an alternative explanation:
RFF ridgeless regression $\approx$ kernel smoothing $\approx$ similarity-weighted average of past returns $\approx$ volatility-timed momentum.

Motivation II: What Needs to Be Explained?

1. High oos Sharpe ratios do not by themselves reveal what the model has learned.
2. The key question is whether RFF learns a genuine nonlinear return-predictability function, or whether the estimator mechanically imposes a trading rule.
3. When $P > T$, ridgeless regression can be rewritten as weighted averages of past returns.
 - Which may generate a **volatility-timed momentum** strategy without learning a true predictive relation.
4. Why is this important?
 - It is related to the effective boundary of machine learning in asset pricing;
 - It is related to the interpretability of financial machine learning;

Research Question and Hypothesis

Main Question

This paper asks whether the apparent virtue of complexity is actually a mechanical consequence of repeating the same strategy?

Question	Hypothesis
RQ1: What does the model compute?	When $P \gg T$, ridgeless RFF forecasts should reduce to similarity-weighted averages of past training-window returns.
RQ2: What explains performance?	With persistent and heteroskedastic predictors, these weights should resemble volatility-timed momentum and explain most RFF abnormal returns.
RQ3: Does the model learn?	If the strategy is mechanical, it should keep producing momentum-like positions even when returns exhibit reversals or predictor information is largely destroyed.

Contribution

1. Literature on complexity in return prediction

Prior: Using overparameterized models to argue that complexity improves return prediction.¹

Extend: The most striking short-window RFF result reflects a mechanical volatility-timed momentum structure, not learned nonlinear predictability.

2. Literature on kernel methods in asset pricing

Prior: RFF and kernel methods approximate high-dimensional nonlinear transformations through low-dimensional kernel evaluations.²

Extend: Using kernel ridgeless representation to make the RFF strategy economically transparent.

3. Literature on limits of learning in empirical asset pricing

Prior: Examining model complexity, measurement error, and high-dimensional learning limits.³

Extend: Linking small T , large P , predictor persistence, heteroskedasticity, and noise to a concrete failure mode in return prediction.

¹Belkin et al. (2019); Bartlett et al. (2020); Hastie et al. (2022); Kelly, Malamud, and Zhou (2024).

²Rahimi and Recht (2007); Sutherland and Schneider (2015); Kozak (2023).

³Berk (2023); Buncic (2025); Cartea, Jin, and Shi (2025); Fallahgoul (2025).

Ridgeless regression with $P > T$

$$\hat{b}_t = (Z'_{t-1} Z_{t-1})^+ Z'_{t-1} r_t, \quad (1)$$

$$P > T$$



ridgeless regression has an interpolating minimum-norm solution



$$\hat{r}_{t+1|t}^{\text{rff}} = z'_t \hat{b}_t = w'_t r_t$$



$$w'_t = z'_t Z'_{t-1} (Z_{t-1} Z'_{t-1})^{-1}$$



w_t = similarity-based weights on past returns



persistent predictors $\Rightarrow$ recent observations are most similar

$\Rightarrow$ recent returns receive high weights $\Rightarrow$ momentum-like forecast.

RFF Forecasts as Kernel-Based Similarity Forecasts

1. RFF transforms a small predictor vector into many nonlinear features.

$$x_t \in \mathbb{R}^K, \quad K = 15 \quad \implies \quad z_t \in \mathbb{R}^P, \quad P = 12,000.$$

$$z'_t z_{t-k} \approx k(x_t, x_{t-k}) = \exp\left(-\frac{\gamma^2}{2} \|x_t - x_{t-k}\|_2^2\right).$$

2. Therefore, the ridgeless RFF forecast has a kernel representation.

$$\hat{r}_{t+1|t}^{\text{rff}} = z'_t Z'_{t-1} (Z_{t-1} Z'_{t-1})^{-1} r_t \approx k(x_t, X_{t-1}) K(X_{t-1}, X_{t-1})^{-1} r_t.$$

3. KMZ standardize RFF, this manuscript uses a shifted-kernel adjustment.

$$\tilde{z}'_t \tilde{z}_{t-k} = z'_t \Omega_{t-1}^{-1} z_{t-k}, \quad \text{so the simple kernel approximation is not exact.}$$

$$\hat{r}_{t+1|t}^{\text{kernel}} \approx [\rho \bar{k}(x_t, X_{t-1}) \iota' + k(x_t, X_{t-1})] K(X_{t-1}, X_{t-1})^{-1} r_t, \quad \rho = 0.69.$$

Takeaway: RFF complexity can be interpreted as kernel smoothing: the forecast weights past returns according to similarity between current and past predictor states.

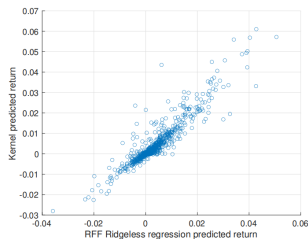
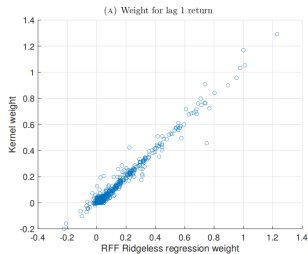
Test of H1: RFF Forecasts as Similarity-Weighted Returns

The weight of RFF on lag 1 return is almost identical to the weight of shifted kernel, with a correlation coefficient of 0.99;

RFF predicted returns are very similar to kernel approach predicted returns, with a correlation coefficient of 0.93.

- RFF forecast is a weighted linear combination of the T training returns.
- Weights are governed by similarity between current and lagged predictor vectors.

Takeaway. H1 is supported. Standardized RFF forecasts and weights are closely approximated by the shifted-kernel representation.



Spanning Tests

Step 1: approximate RFF

- Compare RFF ridgeless regression with shifted kernel ridgeless regression.
- Examine whether forecasts and return weights are nearly identical.

Step 2: build a simple benchmark

$$\hat{r}_{t+1|t}^{\text{volmom}} = 0.05 \frac{1}{\hat{\sigma}_{x,t-1}^2} \sum_{k=0}^{11} \frac{12-k}{78} r_{t-k+1} \quad (2)$$

Step 3: spanning tests

$$r_t^{\text{RFF}} = a + \beta_M r_t^M + \varepsilon_t \quad (3)$$

$$r_t^{\text{RFF}} = a + \beta_M r_t^M + \beta_S r_t^S + \varepsilon_t \quad (4)$$

- $S \in \{\text{Kernel}, \text{VolMom}\}$.
- If RFF alpha disappears, the complex strategy is spanned by a simpler strategy.

Test of H2: Volatility-Timed Momentum Explains RFF Performance

	High-Complexity RFF	Kernel	Vol-Timed Momentum
Panel A: One-Factor α (Market Factor)			
Alpha	0.034	0.040	0.034
(t-stat.)	(2.417)	(2.900)	(3.684)
Information Ratio	0.255	0.306	0.388
Panel B: Two-Factor α (Market and Kernel Factors)			
Alpha	-0.001	—	—
(t-stat.)	(-0.122)	—	—
Information Ratio	-0.013	—	—
Panel C: Two-Factor α (Market and Vol-Timed Momentum Factors)			
Alpha	0.012	—	—
(t-stat.)	(0.945)	—	—
Information Ratio	0.100	—	—

The spanning tests support H2: once the kernel or volatility-timed momentum factor is added, the RFF alpha becomes economically small and statistically insignificant.

Counterfactual Tests

Artificial returns with reversals

- Modify the dependent variable.
- Add an MA(2) component with negative autocorrelation.
- Test whether the RFF strategy adapts away from momentum.

$$\tilde{r}_t = r_t + \xi_t - \theta_1 \xi_{t-1} - \theta_2 \xi_{t-2} \quad (5)$$

Takeaway: A genuinely learned strategy should respond to these changes; a mechanical strategy should continue to produce the same type of weights.

Artificial wild-bootstrap predictors

- Modify the predictors.
- Destroy much of their predictive information.
- Preserve persistence and heteroskedasticity.

$$\tilde{x}_{k,t} = \hat{\phi}_{0,k} + \hat{\phi}_{1,k} \tilde{x}_{k,t-1} + \hat{\varepsilon}_{k,t} \eta_{k,t} \quad (6)$$

Test of H3: RFF Does Not Adapt to Counterfactual Data

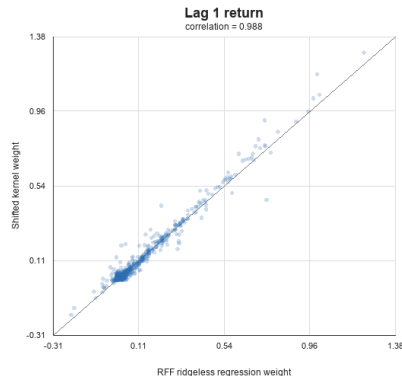
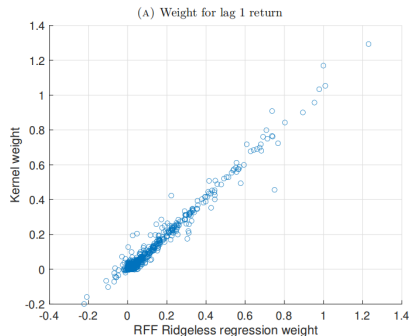
	High-Complexity RFF	Kernel	Vol-Timed Momentum
Artificial returns with reversals			
Alpha	-0.143	-0.137	-0.124
(t-stat.)	(-1.835)	(-1.757)	(-4.099)
Information Ratio	-0.193	-0.185	-0.432
Artificial wild-bootstrap predictors			
Alpha	0.025	0.030	0.034
(t-stat.)	(2.161)	(2.640)	(3.684)
Information Ratio	0.228	0.278	0.388
RFF alpha after controlling for Vol-Timed Momentum			
Artificial returns with reversals	-0.045	—	—
Artificial wild-bootstrap predictors	0.008	—	—

When returns are modified to exhibit reversals, the RFF strategy performs poorly rather than adapting;

When predictor information is largely destroyed but persistence and heteroskedasticity are preserved, performance remains close to the original case.

This supports H3: the strategy behaves mechanically rather than learning whether momentum or reversal is present.

Replication: Figure 2



- 原文 Figure 2 的核心含义：高复杂度 RFF 在短窗口下给最近收益的权重，与 shifted Gaussian kernel 权重几乎一致。
- 使用文章 1 官方数据，设定 $T = 12$, $P = 12000$, $\gamma = 2$ ，对 1000 个 random-feature seeds 取平均。
- 复现结果：原文（左图）的 lag 1 权重相关性为 0.99；复现（右图）结果为 0.988。

Takeaway. Figure 2 的机制结论被严格复现：所谓复杂模型的短窗口预测，权重结构上几乎等价于 shifted kernel/volatility-timing 型规则。

Replication: Table 1

Panel	Model	Our replication			Paper		
		Alpha	t	IR	Alpha	t	IR
A	High-Complexity RFF	0.034	2.249	0.239	0.034	2.417	0.255
A	Kernel	0.040	2.715	0.289	0.040	2.900	0.306
A	Vol-Timed Momentum	0.034	3.570	0.380	0.034	3.684	0.388
B	RFF α vs. Market+Kernel	-0.001	-0.144	-0.015	-0.001	-0.122	-0.013
C	RFF α vs. Market+VolMom	0.012	0.849	0.091	0.012	0.945	0.100

- 与原文相比，alpha、 t 值和 information ratio 均非常接近。
- Panel B 显示 Kernel 因子几乎完全解释 RFF alpha; Panel C 显示 Vol-Timed Momentum 也解释了相当一部分 RFF 表现。

Takeaway. Table 1 的数值结论被成功复现：RFF 的市场择时表现并非独立的新非线性可预测性，而主要来自 kernel/volatility-timing 机制。

China Market Extension: Data and Design

- 数据来源：CSMAR A 股个股月收益、日收益、无风险利率，以及 94 个股票特征文件。
- 市场收益：使用沪深/创业板/科创板/北交所 A 股，以上月总市值加权构造月度市场收益，并减去 CSMAR 月度无风险利率得到市场超额收益。
- 波动率变量：用日度个股收益和滞后市值先构造日度市场收益，再按月计算 realized variance，并滞后一月进入预测变量。
- 主规格：15 个预测变量，包括 lagged market return、lagged realized variance、13 个市值加权股票特征聚合变量⁴；样本期为 1997–2024。
- 稳健性规格：32 个预测变量，包括 30 个股票特征聚合变量。

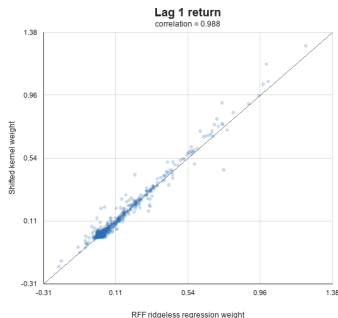
Takeaway. 中国市场扩展不是照搬美国宏观变量，而是在同一机制框架下检验：当预测变量换成中国 A 股特征聚合变量后，RFF 的权重结构是否仍呈现 Nagel (2025) 指出的 kernel/volatility-timing 特征。

⁴03_beta, 06_idvol, 07_vol, 08_idskew, 09_skew, 11_turn, 13_volumed, 15_illiq, 16_LM, 22_mom12, 23_mom6, 24_mom36, 28_bm

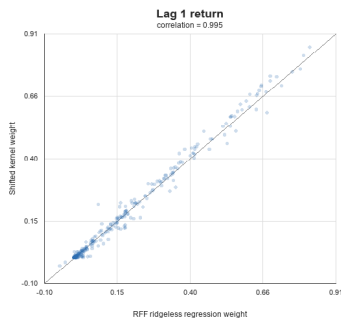
Replication: China Market Extension

Spec.	Model	Alpha	t	IR	Weight corr.
15 predictors	High-Complexity RFF	-0.036	-0.641	-0.125	0.995 / 0.987
15 predictors	Kernel	-0.059	-1.017	-0.198	—
15 predictors	Vol-Timed Momentum	0.036	1.085	0.211	—
32 predictors	High-Complexity RFF	0.035	0.949	0.222	0.999 / 0.986
32 predictors	Kernel	0.030	0.868	0.203	—
32 predictors	Vol-Timed Momentum	0.086	2.193	0.513	—

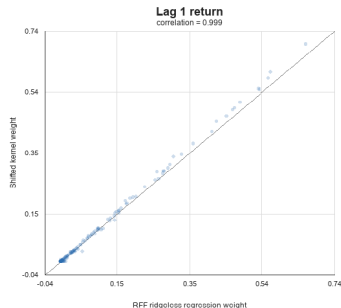
- 权重相关性为 Figure 2 风格的 lag 1 / lag 2 RFF–Kernel 权重相关性。
- 中国市场中，RFF 与 Kernel 权重仍高度一致，机制结果非常稳健。
- 但绩效结果弱于原文美国样本：主规格中 RFF alpha 不显著；扩展规格中 Vol-Timed Momentum 更强，RFF 独立 alpha 仍不显著。



(a) 原文



(b) 15 变量



(c) 32 变量

Takeaway. 中国市场没有简单复制美国样本的强 alpha，但清楚复制了 Figure 2 的权重机制；这表明本文的“复杂模型看似有效”机制具有跨市场稳健性，而收益显著性依赖具体市场环境。

中国市场扩展：结果解读与下一步

目前结果

- **机制结果较强**：在中国 A 股数据中，RFF 权重与 shifted kernel 权重仍高度一致。
 - 15 predictors: lag 1 / lag 2 权重相关性为 0.995/0.987。
 - 32 predictors: lag 1 / lag 2 权重相关性为 0.999/0.986。
- **收益结果较弱**：中国市场没有稳定复现美国样本中 RFF 的显著正 alpha。
 - 15 predictors: RFF alpha = -0.036 , $t = -0.641$ 。
 - 32 predictors: RFF alpha = 0.035 , $t = 0.949$ 。
- **Volatility-timed momentum 更突出**：在 32 predictors 设定下，Vol-Timed Momentum 的 alpha 为 0.086 , $t = 2.193$ ，强于 RFF 与 Kernel。

下一步尝试

- 分市场样本：主板-only、主板 + 创业板、全 A 股，考察市场结构差异。
- 改变市场组合构造：总市值加权、流通市值加权、等权收益。
- 按经济含义分组特征：动量、波动率/流动性、估值、盈利质量、投资增长。

讨论

1. **重要性：**它探讨了高复杂性机器学习是否真的能在资产市场中学习回报的可预测性。
2. **问题本身：**它侧重于一个引人注目的实证主张，其中 $P=12,000$ $T=12$ 的观测值。
3. **经济透明度：**它将一个看似复杂的 RFF 估计器转换为一个简单的核平滑和波动定时动量机制。
4. **分层证据：**数学表示、收益测试、人工反事实测试都指向相同的解释。
5. **平衡性：**该论文总体上不拒绝复杂性；它拒绝了小样本显示出强学习非线性可预测性的说法。

附录：为什么使用 RFF，而不是其他复杂机器学习模型？

$$x_t \in \mathbb{R}^K \xrightarrow{\text{RFF}} z_t \in \mathbb{R}^P, \quad P \gg K, P \gg T$$

- **可系统增加复杂度：** RFF 将少数原始预测变量扩展为大量非线性特征，用于捕捉潜在的非线性关系和交互效应。
- **具有明确的核方法解释：** 当 P 足够大时，RFF 内积近似 Gaussian kernel，因此模型机制可以被数学拆解。
- **复杂度控制更干净：** 研究者可以直接通过 P 调节模型复杂度，便于检验“复杂性是否带来样本外收益”。
- **保留线性回归框架：** RFF 只改变特征空间，估计仍是 ridge/ridgeless regression，便于推导、复现和与传统预测回归比较。

核心优势：RFF 既足够复杂，又足够可解释。

无岭估计与预测分解

在 $P > T$ 时，先把高维无岭回归的预测值从“estimated coefficients” 改写为“weighted past returns”。

1. Ridgeless OLS estimator

$$\hat{b}_t = (Z'_{t-1} Z_{t-1})^+ Z'_{t-1} r_t, \quad \text{其中:} \quad Z_{t-1} \in \mathbb{R}^{T \times P}, \quad r_t \in \mathbb{R}^{T \times 1}, \quad \hat{b}_t \in \mathbb{R}^{P \times 1}.$$

当 $P > T$ 时， $Z'_{t-1} Z_{t-1}$ 通常不可逆，因此使用 Moore–Penrose 伪逆。这个估计量是在所有能够拟合训练样本的解中，选择范数最小的 minimum-norm interpolating solution。

2. Forecast as a weighted combination of past returns

$$\hat{r}_{t+1|t}^{\text{rff}} = z'_t \hat{b}_t = z'_t (Z'_{t-1} Z_{t-1})^+ Z'_{t-1} r_t.$$

Define $w'_t = z'_t (Z'_{t-1} Z_{t-1})^+ Z'_{t-1}$. Then $\hat{r}_{t+1|t}^{\text{rff}} = w'_t r_t$

预测值表面上来自高维系数 $\hat{b}_t$ ，但代入估计量后可以写成过去 T 个训练收益的加权线性组合。

3. Equivalent weight representation when $P > T$ If Z_{t-1} has full row rank,

$$(Z'_{t-1} Z_{t-1})^+ Z'_{t-1} = Z'_{t-1} (Z_{t-1} Z'_{t-1})^{-1}.$$

$$w'_t = z'_t Z'_{t-1} (Z_{t-1} Z'_{t-1})^{-1}$$

这个等价式把权重写成“当前预测状态—过去预测状态”的内积，再乘以训练窗口内状态相似度矩阵的逆：它是后续相似度解释的关键。

拆解 w'_t

1. Z_{t-1} 收集训练窗口内的过去预测变量状态。

$$Z_{t-1} = \begin{pmatrix} z'_{t-1} \\ z'_{t-2} \\ \vdots \\ z'_{t-T} \end{pmatrix} \in \mathbb{R}^{T \times P}, \quad Z'_{t-1} = \begin{pmatrix} z_{t-1} & z_{t-2} & \cdots & z_{t-T} \end{pmatrix} \in \mathbb{R}^{P \times T}.$$

2. 这一项衡量当前预测状态 z_t 与训练窗口中每一个过去状态的相似度。如果 $z'_t z_{t-k}$ 较大, 说明当前状态更接近第 k 个滞后状态。

$$z'_t Z'_{t-1} = \begin{pmatrix} z'_t z_{t-1} & z'_t z_{t-2} & \cdots & z'_t z_{t-T} \end{pmatrix} \in \mathbb{R}^{1 \times T}.$$

3. $Z_{t-1} Z'_{t-1}$ 衡量过去预测状态之间的相似度。

$$Z_{t-1} Z'_{t-1} = \begin{pmatrix} z'_{t-1} z_{t-1} & z'_{t-1} z_{t-2} & \cdots & z'_{t-1} z_{t-T} \\ z'_{t-2} z_{t-1} & z'_{t-2} z_{t-2} & \cdots & z'_{t-2} z_{t-T} \\ \vdots & \vdots & \ddots & \vdots \\ z'_{t-T} z_{t-1} & z'_{t-T} z_{t-2} & \cdots & z'_{t-T} z_{t-T} \end{pmatrix} \in \mathbb{R}^{T \times T}.$$

其逆矩阵用于校正过去状态之间的相互重叠和冗余, 使 w_t 不是简单内积, 而是回归意义上的相似度权重。

w_t 的经济含义

1. $w_{k,t}$ 是模型在预测下一期收益时分配给第 k 个滞后收益的权重。因此，高复杂度预测值可以被理解为过去收益的相似度加权组合。

$$w'_t = z'_t Z'_{t-1} (Z_{t-1} Z'_{t-1})^{-1} = (w_{1,t} \quad w_{2,t} \quad \cdots \quad w_{T,t}).$$

$$r_t = \begin{pmatrix} r_t \\ r_{t-1} \\ \vdots \\ r_{t-T+1} \end{pmatrix}.$$

$\hat{r}_{t+1|t}^{\text{rff}} = w'_t r_t = w_{1,t} r_t + w_{2,t} r_{t-1} + \cdots + w_{T,t} r_{t-T+1}$

2. 过去回报的相似性加权平均值

$w_{k,t}$ is high when z_{t-k} is similar to z_t .

Current state $z_t \approx$ past state $z_{t-k} \implies r_{t-k+1}$ receives a larger weight.

如果当前市场状态与某个过去月份的预测状态相似，模型就更重视那个过去月份之后的收益。

3. 在短窗口中，预测变量高度持续会使最近状态最像当前状态，因此最近收益获得更高权重

Persistent predictors $\Rightarrow z_{t-1}, z_{t-2}$ are usually closer to z_t than distant lags.

$\Rightarrow w_{1,t}, w_{2,t}$ tend to be larger $\Rightarrow \hat{r}_{t+1|t}^{\text{rff}}$ puts more weight on recent returns.

$P > T \implies$ weighted past returns $\implies$ momentum-like forecast under persistent predictors.

RFF-Kernel 近似 ($z'_t z_{t-k} \approx k(x_t, x_{t-k})$) 推导

1. **RFF** 将原始预测变量映射到高维非线性特征空间 $x_t \in \mathbb{R}^K$, $K = 15$, $z_t \in \mathbb{R}^P$, $P = 12,000$.

$$z_t = \sqrt{\frac{2}{P}} \begin{pmatrix} \cos(\gamma \omega'_1 x_t) \\ \sin(\gamma \omega'_1 x_t) \\ \vdots \\ \cos(\gamma \omega'_{P/2} x_t) \\ \sin(\gamma \omega'_{P/2} x_t) \end{pmatrix}, \quad \omega_i \sim IID N(0, I).$$

2. 利用三角恒等式展开两个 **RFF** 向量的内积, 令 $s = t - k$, 则

$$z'_t z_s = \frac{2}{P} \sum_{i=1}^{P/2} [\cos(\gamma \omega'_i x_t) \cos(\gamma \omega'_i x_s) + \sin(\gamma \omega'_i x_t) \sin(\gamma \omega'_i x_s)].$$

由 $\cos a \cos b + \sin a \sin b = \cos(a - b)$, 可得 $z'_t z_s = \frac{2}{P} \sum_{i=1}^{P/2} \cos(\gamma \omega'_i (x_t - x_s))$.

3. 当 P 很大时, 样本平均近似其期望

$$z'_t z_s \approx \mathbb{E}_\omega [\cos(\gamma \omega' (x_t - x_s))].$$

由于 $\omega \sim N(0, I)$, 正态分布的特征函数给出

$$\mathbb{E}_\omega [\cos(\gamma \omega' (x_t - x_s))] = \exp\left(-\frac{\gamma^2}{2} \|x_t - x_s\|_2^2\right).$$

因此, 高维 **RFF** 内积可以被解释为原始 K 维预测变量之间的 **Gaussian kernel** 相似度. $z'_t z_{t-k} \approx k(x_t, x_{t-k}) = \exp\left(-\frac{\gamma^2}{2} \|x_t - x_{t-k}\|_2^2\right)$.

标准化 RFF 与带平移修正的核近似

问题: **KMZ** 实际使用的不是原始 **RFF**, 而是窗口内标准化后的 **RFF**。

$$\tilde{z}_{s,j} = \frac{z_{s,j}}{\hat{\sigma}_{j,t-1}}, \quad s = t-1, \dots, t-T. \quad \text{其中 } \hat{\sigma}_{j,t-1} \text{ 是第 } j \text{ 个 RFF 在当前训练窗口内的标准差。}$$

1. 标准化改变了 **RFF** 内积。原始 **RFF** 的内积是: $z_t' z_{t-k}$, 标准化后变成: $\tilde{z}_t' \tilde{z}_{t-k} = z_t' \Omega_{t-1}^{-1} z_{t-k}$,

其中 $\Omega_{t-1} = \frac{1}{T-1} \text{diag}(Z_{t-1}' Z_{t-1})$ 是一个 $P \times P$ 对角矩阵, 对角线元素为各个 **RFF** 在训练窗口内的方差。

2. 因此, 普通 **Gaussian-kernel** 近似不再直接适用

未标准化时: $z_t' z_{t-k} \approx k(x_t, x_{t-k})$ 。

但标准化后: $\tilde{z}_t' \Omega_{t-1}^{-1} z_{t-k} \not\approx k(x_t, x_{t-k})$ in the same direct way.

3. **Nagel** 的经验修正: 带平移修正的核近似

Nagel 用 shifted kernel ridgeless regression 近似标准化 **RFF** 的预测:

$$\hat{r}_{t+1|t}^{\text{kernel}} \approx \left[\rho \bar{k}(x_t, X_{t-1}) \iota_T' + k(x_t, X_{t-1}) \right] K(X_{t-1}, X_{t-1})^{-1} r_t, \quad \rho = 0.69.$$

其中:

$$k(x_t, X_{t-1}) = (k(x_t, x_{t-1}), \dots, k(x_t, x_{t-T})),$$

$$\bar{k}(x_t, X_{t-1}) = \frac{1}{T} k(x_t, X_{t-1}) \iota_T', \quad \iota_T = (1, \dots, 1)'.$$

结论: 标准化使 **RFF** 内积从普通 **dot product** 变为 **weighted dot product**。

虽然没有闭式 **kernel** 结果, 但在 **KMZ** 设定中, 带平移修正的核近似可以很好复制标准化 **RFF** 的权重和预测。

预测变量持续性产生动量型权重

核心问题：为什么最近收益获得更高权重？

$$k(x_t, x_{t-k}) = \exp\left(-\frac{\gamma^2}{2} \|x_t - x_{t-k}\|_2^2\right) \quad (7)$$

- Gaussian kernel 衡量当前预测变量 x_t 与历史预测变量 x_{t-k} 的相似度。
- Goyal–Welch 预测变量高度持续，因此在短窗口 $T = 12$ 中，最近的预测变量通常最接近当前状态。
- 因而：

$$x_t \approx x_{t-1} \Rightarrow k(x_t, x_{t-1}) \text{ 较大} \Rightarrow r_t \text{ 获得较高权重.}$$

- 结果是：RFF/kernel 预测值更依赖近期收益，形成动量型预测。

Takeaway. RFF 策略不是主动学到动量，而是短窗口中的相似度机制机械地提高了近期收益权重。

预测变量波动率产生择时效应

核心问题：为什么低预测变量波动率时期头寸更大？

$$\|x_t - x_{t-k}\|_2^2 \downarrow \Rightarrow k(x_t, x_{t-k}) \uparrow \Rightarrow \text{过去收益权重上升} \quad (8)$$

- 当训练窗口内预测变量波动率较低时， x_t 与历史 x_{t-k} 的距离较小。
- 距离越小，Gaussian kernel 相似度越高，过去收益的整体权重越大。
- 当预测变量波动率较高时，距离扩大，相似度下降，策略头寸收缩。
- 因此，RFF/kernel 策略表现为：

低预测变量波动率 $\Rightarrow$ 高权重/高头寸.

Takeaway. RFF 策略的 volatility timing 主要来自 predictor volatility，而不是模型真正学到了收益波动率择时规律。

Volatility-Timed Momentum 的例子

设当前为 2020 年 12 月，预测 2021 年 1 月市场超额收益。

$$\hat{r}_{t+1|t}^{\text{volmom}} = 0.05 \times \frac{1}{\hat{\sigma}_{x,t-1}^2} \times \left(\frac{12}{78} r_t + \frac{11}{78} r_{t-1} + \cdots + \frac{1}{78} r_{t-11} \right)$$

- 时序动量：若过去几个月市场指数持续上涨，

$$r_t, r_{t-1}, r_{t-2} > 0,$$

且最近收益权重最高，则动量信号为正，模型提高市场暴露。

- 波动率择时：若 15 个预测变量在过去 12 个月内波动较低，

$$\hat{\sigma}_{x,t-1}^2 \downarrow \Rightarrow \frac{1}{\hat{\sigma}_{x,t-1}^2} \uparrow,$$

则同样的正动量信号会被进一步放大。

- 合成结果：市场近期上涨决定“做多方向”，预测变量低波动决定“更大仓位”。

Takeaway. VolMom 策略不是简单追涨，而是在近期市场动量信号上叠加预测变量波动率择时。